PAYLOAD BUS BREAKOUT

ARK Pixhawk Payload Bus Breakout

Pixhawk Payload Bus to Pixhawk-standard JST-GH, with a protected 5 V peripheral rail

The USA-built ARK Pixhawk Payload Bus Breakout fans the 30-pin Payload Bus out to individually keyed Pixhawk-standard JST-GH connectors. UART4/I2C3, CAN, Ethernet, USB, and GPIO each get their own connector, so a payload can be wired to an ARK carrier without a custom harness.

A 5 V supply enters on a 6-pin Molex Pico-Clasp using the Pixhawk power-module pinout. An onboard TI BQ24315 protection switch derives the VDD_5V_PERIPH rail with a 1.5 A current limit, overvoltage protection, and a fault flag, and is gated by the autopilot over VDD_5V_PERIPH_nEN.

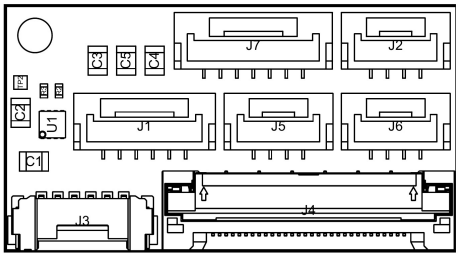
KEY FEATURES

- 30-pin, 0.5 mm pitch Payload Bus FFC connector
- UART4 and I2C3 on 6-pin JST-GH
- CAN on 4-pin JST-GH
- Ethernet on 4-pin JST-GH
- USB on 4-pin JST-GH
- GPIO, FMU PWM, and capture on 7-pin JST-GH
- 1.5 A protected 5 V peripheral rail (TI BQ24315)
- Overvoltage and overcurrent protection with fault flag
- Test points for VDD_5V_PERIPH, FAULT, and VDD_5V_HIGHPOWER_nEN
- Includes 30-pin, 0.5 mm pitch, 150 mm FFC cable

SPECIFICATIONS

ELECTRICAL	
Supply voltage	5 V (J3)
Peripheral rail	VDD_5V_PERIPH, 1.5 A current limit
Rail enable	VDD_5V_PERIPH_nEN (active low)
Protection	Overvoltage, overcurrent, fault flag
Signal logic level	3.3 V
Operating temperature	−25 °C to +85 °C

MECHANICAL	
Dimensions	34.25 × 18.50 × 5.81 mm
Weight	3.5 g
PCB	2-layer FR-4, 1.61 mm, ENIG



AT A GLANCE	
Payload bus	30-pin FFC
Interfaces	UART, I2C, CAN, Ethernet, USB, GPIO
Peripheral rail	5 V @ 1.5 A
Supply	5 V
Dimensions	34.25 × 18.5 × 5.81 mm
Weight	3.5 g
Hardware rev	Rev 1.0

COMPLIANCE & SOURCING

› Designed and manufactured in the USA

ORDERING & RESOURCES

STORE arkelectron.com/product/ark-pixhawk-payload-bus-breakout/

DOCS docs.arkelectron.com/products/accessories/ark-pixhawk-payload-bus-breakout

CAD github.com/ARK-Electronics/ark-hardware

PAYLOAD BUS CONNECTOR

J4 — 30-pin, 0.5 mm pitch FFC (Amphenol F33M1B7F121030E200)

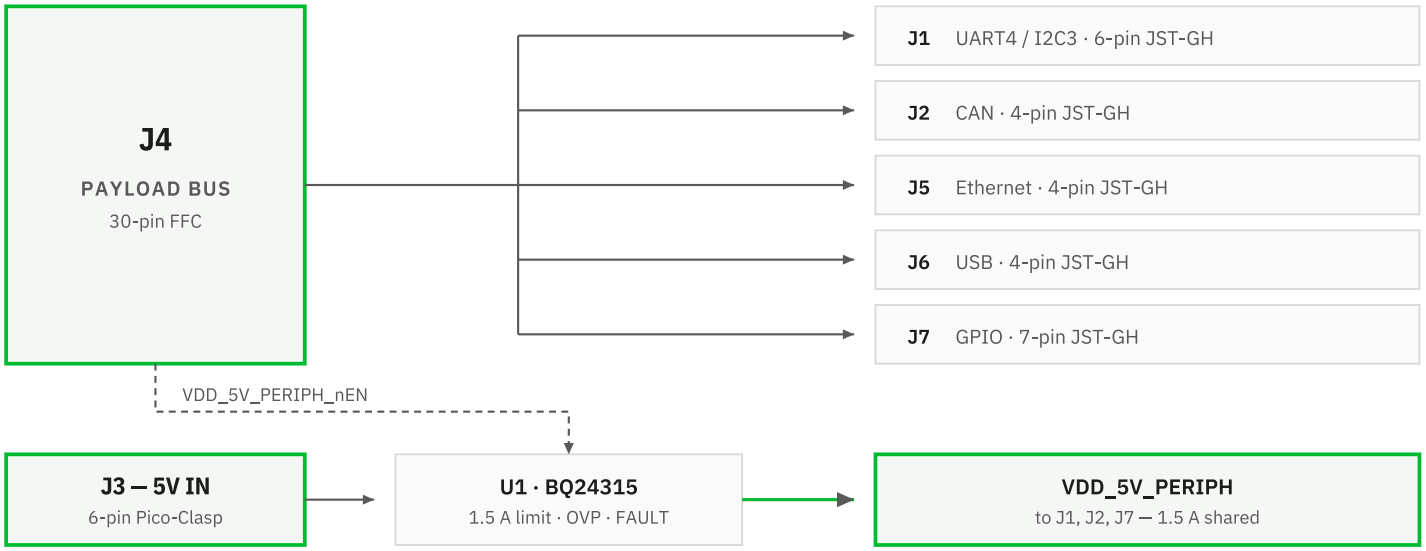
PIN	SIGNAL	LEVEL
1	GND	GND
2	UART4_TX_EXT	3.3V
3	UART4_RX_EXT	3.3V
4	GND	GND
5	I2C3_SDA_EXT	3.3V
6	I2C3_SCL_EXT	3.3V
7	GND	GND
8	CAN_P	5V
9	CAN_N	5V
10	GND	GND
11	FMU_CH7_EXT	3.3V
12	GPIO12_EXT	3.3V
13	FMU_CH8_EXT	3.3V
14	GND	GND
15	FMU_CAP_EXT	3.3V
16	GND	GND

PINS 17–30

PIN	SIGNAL	LEVEL
17	PYLD_ETH_TX_P	Diff.
18	PYLD_ETH_TX_N	Diff.
19	GND	GND
20	PYLD_ETH_RX_P	Diff.
21	PYLD_ETH_RX_N	Diff.
22	GND	GND
23	GPIO01	3.3V
24	VBUS	5V
25	USB_N	3.3V
26	USB_P	3.3V
27	GND	GND
28	VDD_5V_HIGHPOWER_nEN	3.3V
29	VDD_5V_PERIPH_nEN	3.3V
30	GND	GND
G1–G6	GND	GND

Pin 28 (VDD_5V_HIGHPOWER_nEN) is brought out to test point TP1 only and is not otherwise connected on this board. Pin 29 (VDD_5V_PERIPH_nEN) drives the enable input of the onboard protection switch through a 47 kΩ series resistor; pulling it low enables the VDD_5V_PERIPH rail.

BLOCK DIAGRAM



J5 and J6 carry no supply rail. VBUS on J6 is passed through directly from Payload Bus pin 24.

BREAKOUT CONNECTORS

J3 — 5 V input, 6-pin Pico-Clasp

PIN	SIGNAL	LEVEL
1	5V	5V
2	5V	5V
3	I2C3_SCL_EXT	3.3V
4	I2C3_SDA_EXT	3.3V
5	GND	GND
6	GND	GND

J1 — UART4 / I2C3, 6-pin JST-GH

PIN	SIGNAL	LEVEL
1	VDD_5V_PERIPH	5V
2	UART4_TX_EXT	3.3V
3	UART4_RX_EXT	3.3V
4	I2C3_SCL_EXT	3.3V
5	I2C3_SDA_EXT	3.3V
6	GND	GND

J2 — CAN, 4-pin JST-GH

PIN	SIGNAL	LEVEL
1	VDD_5V_PERIPH	5V
2	CAN_P	5V
3	CAN_N	5V
4	GND	GND

J5 — Ethernet, 4-pin JST-GH

PIN	SIGNAL	LEVEL
1	PYLD_ETH_RX_N	Diff.
2	PYLD_ETH_RX_P	Diff.
3	PYLD_ETH_TX_N	Diff.
4	PYLD_ETH_TX_P	Diff.

J6 — USB, 4-pin JST-GH

PIN	SIGNAL	LEVEL
1	VBUS	5V
2	USB_N	3.3V
3	USB_P	3.3V
4	GND	GND

J7 — GPIO, 7-pin JST-GH

PIN	SIGNAL	LEVEL
1	VDD_5V_PERIPH	5V
2	FMU_CH7_EXT	3.3V
3	FMU_CH8_EXT	3.3V
4	FMU_CAP_EXT	3.3V
5	GPIO01	3.3V
6	GPIO12_EXT	3.3V
7	GND	GND

NOTES

- VDD_5V_PERIPH is shared by J1, J2, and J7 and is limited to 1.5 A in total.
- VBUS on J6 is passed through from Payload Bus pin 24 and is not switched or protected on this board.
- J5 carries the Ethernet pairs only; the connector shell is intentionally left unconnected.
- Test points: TP1 = VDD_5V_HIGHPOWER_nEN, TP2 = VDD_5V_PERIPH, TP3 = FAULT (open drain).
- Operating temperature is limited by the JST-GH connectors; all other components are rated wider.

ARK-DS-ARK-PIXHAWK-PAYLOAD-BUS-BREAKOUT | Version 1.0, 2026-10-29 | Compiled from the ARK Pixhawk Payload Bus Breakout Rev 1.0 schematic, the Rev 1.0 bill of materials, and the Rev 1.0 manufacturing documentation. Signal names are as given in the schematic. Specifications are subject to change without notice. © 2026 ARK Electronics LLC.